Graphs 7 — Using graphs to solve equations · Non-linear simultaneous equations

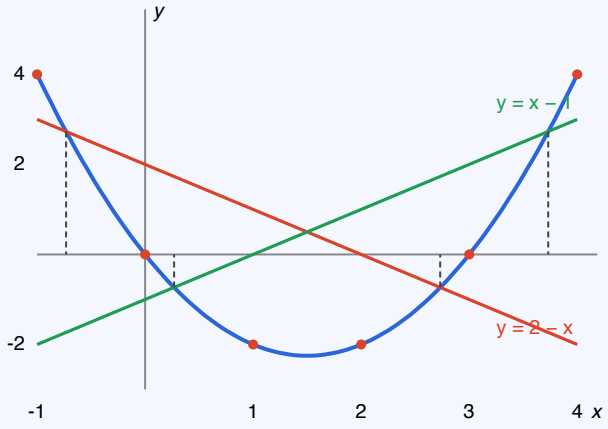
Calculator allowed; graph paper needed. Marks in brackets. ★ = harder question. Rearrange the equation so one side is the curve you have drawn; the other side is the line to draw.

1 **a** Copy and complete the table and draw the graph of $y = x^2 - 3x$ for $-1 \le x \le 4$. [8]
b By drawing suitable straight lines on your graph, solve **i** $x^2 - 3x = 2 - x$ **ii** $x^2 - 4x + 1 = 0$.

x	-1	0	1	2	3	4
y	4		-2			4

a

x	-1	0	1	2	3	4
y	4	0	-2	-2	0	4



b i The left side is the curve already drawn, so draw the line $y = 2 - x$ and read the x-coordinates of the crossing points: $x \approx -0.7$ and $x \approx 2.7$ (exact: $1 \pm \sqrt{3}$).
ii Rearrange so the curve appears: $x^2 - 3x = x - 1$. Draw $y = x - 1$: crossings at $x \approx 0.3$ and $x \approx 3.7$ (exact: $2 \pm \sqrt{3}$).

2 The graph of $y = 2x^2 - 5x + 1$ has been drawn. Find the equation of the straight line that should be drawn to solve each equation. [6]
a $2x^2 - 5x - 3 = 0$ **b** $2x^2 - 7x + 1 = 0$ **c** $2x^2 - 3x - 4 = 0$

Add or subtract to turn the left side into $2x^2 - 5x + 1$, and whatever is left on the right is the line.
a Add 4: $2x^2 - 5x + 1 = 4 \rightarrow$ line $y = 4$. **b** Add $2x$: $2x^2 - 5x + 1 = 2x \rightarrow$ line $y = 2x$. **c** Subtract $2x$ and add 5: $2x^2 - 5x + 1 = 2x + 5 \rightarrow$ line $y = 2x + 5$.

3 The graph of $y = x^3 + 2x^2 - 3x - 1$ has been drawn. Find the equation of the line to draw to solve [4]

a $x^3 + 2x^2 - 3x + 4 = 0$ **b** $x^3 + 2x^2 - 5x - 3 = 0$

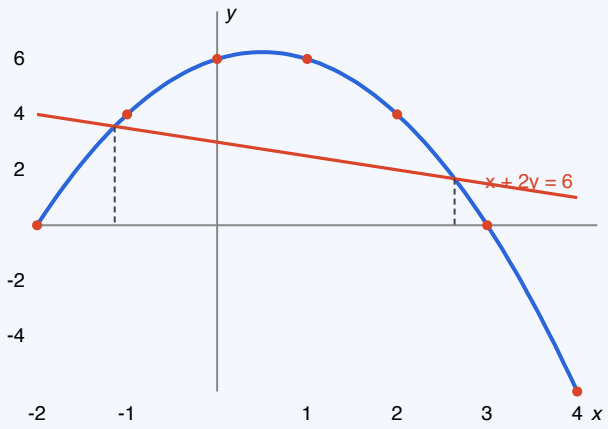
a Subtract 5: $x^3 + 2x^2 - 3x - 1 = -5 \rightarrow$ line $y = -5$. **b** Add $2x + 2$: $x^3 + 2x^2 - 3x - 1 = 2x + 2 \rightarrow$ line $y = 2x + 2$.

4 **a** Copy and complete the table and draw the graph of $y = 6 + x - x^2$ for $-2 \leq x \leq 4$. **b** Use your graph to solve the simultaneous equations $y = 6 + x - x^2$ and $x + 2y = 6$, giving your answers to 1 d.p. [7]

x	-2	-1	0	1	2	3	4
y	0		6		4		-6

a

x	-2	-1	0	1	2	3	4
y	0	4	6	6	4	0	-6



b Rearrange the line: $y = 3 - \dfrac{x}{2}$, through $(0, 3)$ and $(6, 0)$ — draw it on the same axes. The crossing points give both solutions: $x \approx -1.1$, $y \approx 3.6$ and $x \approx 2.6$, $y \approx 1.7$. (Algebra check: $6 + x - x^2 = 3 - \frac{x}{2} \Rightarrow x^2 - 1.5x - 3 = 0$.)

Total: 25 marks

Check: the drawn curve must appear on one side exactly as written — add/subtract the same thing to both sides · the solutions are the x-coordinates where line and curve cross · two crossings → two solutions · for simultaneous equations read both coordinates of each crossing.